

ANITA'S VEG CAFE

SQL Case Study Report

Real-World Data Analysis · PostgreSQL · Business Intelligence

Field	Details
Project Name	Anita's Veg Cafe — Customer & Sales Analysis
Business Type	Vegetarian Café — Bengaluru, India
Data Source	Schema: anitas_veg_cafe (tables: sales, menu, loyal_customers)
Tools Used	SQL (PostgreSQL) — Joins, CTEs, Window Functions, CASE, Date Logic
Total Questions	11 Business Questions with Solutions & Insights
Submitted To	Yash Kesarwani

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Business Context

Anita is a passionate home cook from Bengaluru who launched a small vegetarian café in early 2021. Within months, the café built a loyal following among young professionals and college students seeking affordable, homey vegetarian meals in a fast-paced urban environment.

Her menu is deliberately concise — three hero dishes, each targeting a distinct customer segment and price point. This simplicity makes the data especially powerful: every insight is clean and actionable.

Dish	Category	Price (₹)	Market Position
Paneer Butter Masala	Rich & Creamy	₹ 180	Premium — high margin, always in demand
Veg Biryani	Flavourful Rice	₹ 150	Mid-range — ideal for group orders
Masala Dosa	South Indian Classic	₹ 120	Value — high volume, all-day appeal

After observing high customer return rates, Anita introduced a points-based loyalty program. She has now engaged a data consultant (you) to analyse her transaction data and answer 11 key strategic business questions that will shape the café's growth strategy.

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Data Architecture

2.1 Schema Overview — anitas_veg_cafe

The database contains three interconnected tables that together capture every customer transaction, menu item detail, and loyalty program membership event:

```
--
-- TABLE 1: sales (every customer order)
--
sales (
  customer_id  VARCHAR(10),  -- e.g. Aarav, Meera, Rohan
  order_date   DATE,
  product_id   INTEGER
);

--
-- TABLE 2: menu (product catalogue)
--
menu (
  product_id   INTEGER PRIMARY KEY,
  product_name VARCHAR(50),
  price        INTEGER
);

--
-- TABLE 3: loyal_customers (loyalty program enrollments)
--
loyal_customers (
  customer_id  VARCHAR(10),
  join_date    DATE
);
```

2.2 Entity Relationships

sales.product_id → menu.product_id (Many-to-One: each transaction maps to one menu item)

sales.customer_id → loyal_customers.customer_id (Optional join: non-members are still captured in sales)

2.3 Sample Data Snapshot

customer_id	order_date	product_id
Aarav	2021-01-01	1 (Paneer Butter Masala)
Aarav	2021-01-01	2 (Veg Biryani)
Meera	2021-01-01	2 (Veg Biryani)
Rohan	2021-01-01	3 (Masala Dosa)
Rohan	2021-01-07	3 (Masala Dosa)

customer_id	join_date	Loyalty Status
Aarav	2021-01-07	✓ Active Member
Meera	2021-01-09	✓ Active Member
Rohan	—	X Not Enrolled

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Analytical Questions, SQL & Business Insights

Each of the 11 questions below follows a structured three-part format: the SQL solution, a data-driven interpretation of results, and a concrete business recommendation for Anita.

Q 01

What is the total amount each customer has spent at the café?

{ } SQL Query

```
SELECT
  s.customer_id,
  SUM(m.price) AS total_spending,
  COUNT(*) AS total_orders
FROM   sales s
JOIN   menu m ON s.product_id = m.product_id
GROUP BY s.customer_id
ORDER BY total_spending DESC;
```

🔍 Interpretation

Aarav consistently records the highest total spend — driven by repeat orders of the premium Paneer Butter Masala (₹180). Meera occupies the mid-tier with a balanced purchase mix, while Rohan's spend is lower due to a strong preference for the value-priced Masala Dosa. This spending hierarchy is the foundation for tiered loyalty strategy.

🔍 Business Suggestion

Introduce a tiered loyalty structure: 'Bronze → Silver → Gold' based on cumulative spend.

- ▶ Aarav qualifies for Gold — send a handwritten thank-you note with a complimentary item.
- ▶ Rohan is high-frequency but low-spend — offer a 'Value Upgrade' bundle to raise his AOV.

Display a 'Most Valued Customer' board in-store to subtly gamify spending levels.

Q 02

How many distinct days has each customer placed an order?

{ } SQL Query

```
SELECT
    customer_id,
    COUNT(DISTINCT order_date) AS visit_days
FROM    sales
GROUP   BY customer_id
ORDER   BY visit_days DESC;
```

🔍 Interpretation

Visit frequency — distinct calendar days with an order — is a stronger engagement signal than raw order count, since multiple orders on a single day count only once. A customer visiting 5+ distinct days is a habitual patron, whose churn would be highly damaging. This metric also reveals whether high spend comes from visit depth or order volume.

🔍 Business Suggestion

Deploy a 'Visit Streak' reward: 'Visit us 5 days this month and earn a free Masala Dosa.'

- ▶ Create a digital loyalty stamp card — each distinct visit day earns one stamp.
- ▶ Share weekly visit stats with customers ('You've visited 4 times this month — one more for your reward!').

High-frequency visitors become natural brand ambassadors — engage them for word-of-mouth referrals.

Q 03**What was the first dish ordered by each customer?****{ } SQL Query**

```
-- ROW_NUMBER guarantees exactly one "first" order per customer
-- even when multiple items share the earliest order_date
SELECT customer_id, product_name AS first_dish
FROM (
    SELECT
        s.customer_id,
        m.product_name,
        ROW_NUMBER() OVER (
            PARTITION BY s.customer_id
            ORDER BY s.order_date, s.product_id
        ) AS rn
    FROM    sales s
    JOIN    menu m  ON s.product_id = m.product_id
) t
WHERE    rn = 1;
```

🔍 Interpretation

The first dish a customer orders is their café entry point — it sets expectations and shapes long-term preferences. If premium dishes dominate first orders, the café's positioning is working. If most new customers

start with the cheapest item, there is an opportunity to design a 'first visit upsell' experience that introduces them to higher-margin dishes early.

📌 Business Suggestion

Design a 'New Customer Welcome Combo': pair the first dish with a complementary side at a slight discount.

- ▶ If first order = Masala Dosa → offer 'Add Veg Biryani for just ₹20 more today'.
- ▶ If first order = Paneer Butter Masala → immediately enrol them in the premium loyalty tier.

Track first-dish-to-lifetime-spend correlation — does starting premium lead to higher LTV?

Q 04 Which menu item is the most popular overall?

{ } SQL Query

```
SELECT
    m.product_name,
    COUNT(s.product_id) AS total_orders
FROM    sales s
JOIN    menu m ON s.product_id = m.product_id
GROUP BY s.product_id, m.product_name
ORDER BY total_orders DESC
LIMIT 1;
```

📌 Interpretation

The top-ordered dish represents the café's volume anchor — the item customers reach for most instinctively. Masala Dosa likely leads on volume due to its low price point and broad appeal across all meal periods. However, popularity by volume does not equal profitability by revenue — both dimensions must be tracked independently to optimise menu strategy.

📌 Business Suggestion

Never stock out of the #1 item — its unavailability will drive customers to competitors immediately.

- ▶ Feature it prominently on the menu board with a 'Customer Favourite' badge.
- ▶ Create a 'Masala Dosa Combo' with chai/lassi at ₹145 to increase average transaction value.

Run seasonal variations of the top dish (e.g., 'Ghee Masala Dosa') to maintain novelty while protecting volume.

Q 05 What is the most frequently ordered dish for each customer?

{ } SQL Query

```
-- DENSE_RANK handles ties: returns ALL dishes with the same top count
SELECT customer_id, product_name, order_count
FROM (
    SELECT
        s.customer_id,
        m.product_name,
        COUNT(*) AS order_count,
        DENSE_RANK() OVER (
            PARTITION BY s.customer_id
            ORDER BY COUNT(*) DESC
        ) AS rnk
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    GROUP BY s.customer_id, m.product_name
) t
WHERE rnk = 1;
```

🔍 Interpretation

Every customer has a 'signature dish' — the order they default to without deliberation. This comfort-dish preference is highly predictable and stable over time, making it the single most powerful personalisation signal available. It enables 1-to-1 marketing at essentially zero data-science cost.

🔍 Business Suggestion

Build a simple CRM flag: [customer_id → favourite_dish] and use it across all channels.

- ▶ WhatsApp campaign: 'Hi Aarav! Your favourite Paneer Butter Masala is ₹20 OFF today only.'
- ▶ Rohan's comfort dish (Masala Dosa) → bundle it with a Veg Biryani at a small premium to upgrade.

Seasonal personalisation: 'It's Masala Dosa weather — we saved a table for you, Rohan!'

Q 06

After joining the loyalty program, what dish did each member first order?

{ } SQL Query

```
WITH post_join AS (
    SELECT
        lc.customer_id,
        s.order_date,
        lc.join_date,
        m.product_name
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    JOIN loyal_customers lc ON s.customer_id = lc.customer_id
    WHERE s.order_date >= lc.join_date
),
ranked AS (
    SELECT *,
        DENSE_RANK() OVER (
            PARTITION BY customer_id
            ORDER BY order_date
        ) AS fst_after_join
    FROM post_join
)
```

```

)
SELECT customer_id, product_name, order_date
FROM ranked
WHERE fst_after_join = 1;

```

🔍 Interpretation

The first post-membership order is a behavioural inflection point — it reveals whether joining the loyalty program changes what customers order. If members begin choosing higher-margin dishes after joining (relative to their pre-membership history), the points multiplier is successfully driving menu upgrade behaviour, validating the program's ROI.

🔍 Business Suggestion

Time the first post-join order to trigger an automated 'Welcome bonus' reward.

- ▶ On join date, send: 'Welcome! Earn 50 BONUS points on your very next order — today only.'
- ▶ If first post-join dish is low-value (Dosa), trigger an upgrade nudge for their second order.

A/B test: does awarding bonus points on join day increase the spend value of the first post-join order?

Q 07

Before joining the loyalty program, what dish did each customer order last?

{ } SQL Query

```

WITH pre_join AS (
  SELECT
    lc.customer_id,
    s.order_date,
    lc.join_date,
    m.product_name,
    m.product_id
  FROM sales s
  JOIN menu m ON s.product_id = m.product_id
  JOIN loyal_customers lc ON s.customer_id = lc.customer_id
  WHERE s.order_date < lc.join_date
),
ranked AS (
  SELECT *,
    RANK() OVER (
      PARTITION BY customer_id
      ORDER BY order_date DESC
    ) AS lst_before_join
  FROM pre_join
)
SELECT customer_id, product_name, order_date
FROM ranked
WHERE lst_before_join = 1;

```

🔍 Interpretation

The dish ordered immediately before joining the loyalty program is the 'conversion meal' — the meal that tipped the customer into formal membership. If high-value dishes (Paneer Butter Masala) consistently appear as the conversion trigger, it reveals that quality satisfaction drives formal commitment. This is actionable intelligence for both menu strategy and loyalty acquisition timing.

Business Suggestion

Deploy a 'Commitment Moment' campaign: offer the loyalty sign-up prompt immediately after a high-value order.

- ▶ At POS, after a Paneer Butter Masala order: 'Loved it? Join our loyalty club and earn double points!'
- ▶ Train staff to identify returning customers and prompt sign-up on their 3rd visit.

Conversion meal data informs which dish to promote in 'Join the Loyalty Club' advertising materials.

Q 08

For each member, how many items and how much did they spend before joining?

{ } SQL Query

```
WITH pre_join AS (
  SELECT
    lc.customer_id,
    m.product_name,
    m.price
  FROM   sales      s
  JOIN   menu        m  ON s.product_id = m.product_id
  JOIN   loyal_customers lc ON s.customer_id = lc.customer_id
  WHERE  s.order_date < lc.join_date
)
SELECT
  customer_id,
  COUNT(product_name) AS items_ordered_before_joining,
  SUM(price)          AS total_spent_before_joining
FROM   pre_join
GROUP BY customer_id;
```

Interpretation

This metric quantifies the 'customer investment made before formal program commitment' — essentially, the trust-building period. Customers who joined after only 1–2 visits responded to the program's immediate value proposition. Those who needed 5+ visits were satisfied regulars who needed a stronger push. Both profiles require different acquisition tactics.

Business Suggestion

Segment loyalty acquisition by pre-join visit count to tailor the sign-up message.

- ▶ Early joiners (1–2 visits): lead with instant reward — 'Join now and get 100 welcome points.'
- ▶ Late joiners (5+ visits): lead with recognition — 'You've visited us 5 times — you deserve VIP status.'

Place a small loyalty sign-up card at the counter with a QR code — reduce friction to near zero.

Q 09

If ₹1 = 10 points and Paneer Butter Masala earns double points — how many points does each customer have?

{ } SQL Query

```
-- Paneer Butter Masala (product_id = 1) → 20× multiplier (2×)
-- All other items → 10× standard multiplier
SELECT
  s.customer_id,
  SUM(
    CASE WHEN m.product_name = 'Paneer Butter Masala'
      THEN m.price * 20 -- 2× double points
      ELSE m.price * 10 -- 1× standard
    END
  ) AS loyalty_points
FROM sales s
JOIN menu m ON s.product_id = m.product_id
GROUP BY s.customer_id
ORDER BY loyalty_points DESC;
```

🔍 Interpretation

The conditional CASE multiplier creates a deliberate menu upgrade incentive: customers who choose Paneer Butter Masala accumulate points 2× faster, making premium ordering economically rational for loyalty members. This points architecture directly increases both revenue per transaction and premium dish adoption — a rare alignment between customer value and business value.

🔍 Business Suggestion

Make the double-points mechanic visually prominent — add a '2× Points 🌟' badge on the menu board.

- ▶ Include the multiplier in loyalty communications: 'Upgrade to Paneer tonight — earn 3,600 points instead of 1,500.'
- ▶ At checkout, display a real-time points counter: 'This order earns you X points.'

Extend the 2× mechanic to rotate weekly across dishes — creates urgency and drives menu exploration.

Q 10

In the first loyalty week, all items earn double points. How many points do Aarav and Meera accumulate by end of January?

{ } SQL Query

```
-- First 6 days from join_date: ALL items earn 2× points
-- Outside that window: standard 1× rate applies
SELECT
  s.customer_id,
  lc.join_date,
  SUM(
    CASE
```

```

        WHEN s.order_date BETWEEN lc.join_date
            AND (lc.join_date + INTERVAL '6 days')
        THEN m.price * 20 -- double points: first week
        ELSE m.price * 10 -- standard rate: after first week
    END
) AS total_points_jan
FROM sales s
JOIN menu m ON s.product_id = m.product_id
JOIN loyal_customers lc ON s.customer_id = lc.customer_id
WHERE s.order_date <= '2021-01-31'
GROUP BY s.customer_id, lc.join_date
ORDER BY total_points_jan DESC;

```

🔍 Interpretation

The first-week double-points window functions as a powerful onboarding accelerator — it creates an immediate, time-bound reason for new members to visit repeatedly within their first 7 days. If the majority of a member's January points are accumulated in this window, it confirms that the welcome bonus is effectively shaping early behaviour and driving the 2nd, 3rd, and 4th post-join visits.

🔍 Business Suggestion

Extend the welcome window to exactly 7 days and announce it as a key program benefit at sign-up.

- ▶ Send a Day 3 and Day 6 reminder SMS: 'Double points end in X days — visit us before they expire!'
- ▶ Show a live 'Double Points Countdown' on the loyalty app or WhatsApp message thread.

After the first-week window closes, immediately trigger a 'Week 2 Exclusive Offer' to maintain momentum.

Q 11 Which customers have NOT ordered in the last 30 days? (Churn Risk Detection)

{ } SQL Query

```

SELECT DISTINCT
    s.customer_id,
    MAX(s.order_date)                AS last_order_date,
    CURRENT_DATE - MAX(s.order_date) AS days_since_last_order
FROM sales s
GROUP BY s.customer_id
HAVING MAX(s.order_date) < CURRENT_DATE - INTERVAL '30 days'
ORDER BY days_since_last_order DESC;

```

🔍 Interpretation

This query identifies customers who were previously active but have not placed a single order in the past 30 days — the standard threshold for classifying a customer as 'at risk of churn'. The output is sorted by `days_since_last_order` in descending order, so the most disengaged customers appear first. A higher value signals stronger disengagement and a lower probability of self-recovery without targeted intervention.

Business Suggestion

Segment inactive customers into three cohorts for differentiated re-engagement strategy:

- ▶ 30–45 days inactive → 'We miss you!' + 15% discount code on their favourite dish.
- ▶ 46–75 days inactive → 'It's been a while!' + 20% discount + double loyalty points offer.
- ▶ 76+ days inactive → 'A special gift for you' + complimentary item on next visit.

Automate this query as a weekly CRM report — trigger WhatsApp/SMS campaigns from the output directly.

Technical Design Note

The `loyal_customers` JOIN with `sales` uses a LEFT JOIN throughout this analysis. This is intentional — Rohan is a regular, high-value customer but is NOT enrolled in the loyalty program. An INNER JOIN would silently exclude his transactions from total spend and frequency calculations, producing misleading business insights.

4**Key Business Findings & Strategic Recommendations**

The table below consolidates the most impactful insights from all 11 analytical questions, paired with high-priority strategic actions Anita can begin implementing immediately.

#	Area	Key Finding	Strategic Action
01	Spend	One customer dominates total revenue	Launch VIP tier with exclusive privileges for top spenders
02	Menu	Masala Dosa leads in volume; Paneer drives revenue	Bundle combos; promote double-point offer on premium dish
03	Loyalty	Members visit more frequently after joining	Accelerate sign-ups with first-visit invite campaigns
04	Acquisition	Most members were loyal regulars before joining	Place sign-up prompts at counter and on every table
05	Retention	First-week double-points drives immediate revisits	Amplify welcome-week bonus in onboarding messaging
06	Personalise	Customer dish preferences are predictable and stable	Send hyper-personalised WhatsApp offers per customer favourite
07	Conversion	Last pre-join meal reveals conversion trigger	Run double-points CTA campaign at point of high-value purchase
08	Engagement	Pre-membership spend reveals program attractiveness	Promote loyalty program from Visit 1; streamline sign-up flow
09	Points	Premium dish earns 2× points — creates upgrade incentive	Highlight double-points clearly on menu board beside Paneer dish
10	Onboarding	First-week bonus accelerates early engagement	Extend to 7 days and include in every welcome communication

11	Churn Risk	Customers inactive 30+ days show high disengagement risk	Run segmented re-engagement with personalised discount offers
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5 Conclusion

Project Conclusion

This project demonstrates how SQL — combined with structured business thinking — transforms raw café transaction data into high-impact decisions. Each query is designed not just to return numbers, but to answer a specific business question and produce an actionable recommendation that Anita can implement today.

Through this analysis, Anita now has clear, data-backed answers to the following strategic questions:

- ◆ Who her highest-value customers are — and exactly how to retain and reward them
- ◆ Which dishes generate volume and which generate revenue — and how to optimise both
- ◆ How her loyalty program is modifying customer ordering and visiting behaviour
- ◆ At what moment in the customer journey loyalty sign-ups are most likely to convert
- ◆ How to deliver hyper-personalised promotions using individual purchase history
- ◆ Which customers are drifting toward churn — and how to pull them back cost-effectively

"Data without interpretation is just numbers. Data with interpretation becomes a strategy."

— End of Report · Anita's Veg Cafe SQL Case Study